

Structural Reduction of Two Twin-Prime Conjectures and Conditional Resolution under Hypothesis H

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Abstract

I introduced the Twin-Prime Propagation Conjecture and the Twin-Gap Existence Conjecture in two earlier notes. Here I give a complete structural reduction of both statements. A covering argument shows that the second linear form D_n is never a lower twin prime after the first pair; consequently the propagation conjecture reduces to a single admissible 6-tuple of linear forms. That 6-tuple is a special case of Schinzel's Hypothesis H, so the propagation conjecture follows from Hypothesis H. The gap-existence conjecture is strictly stronger: it is a uniform short-interval form of the Bateman–Horn law. Both statements imply the classical twin-prime conjecture. Neither is accessible to present sieve methods unconditionally. A companion Python certificate (`prove_section6.py`) machine-checks every modular lemma, the admissibility statements, the logical reductions, and the computational claims of Section 6.

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1 The two conjectures

Let $\mathcal{T}$ denote the strictly increasing sequence of lower twin primes: primes p for which $p + 2$ is also prime,

$$3, 5, 11, 17, 29, 41, 59, 71, 101, \dots$$

Write p_n for the n th term.

Conjecture 1.1 (Twin-Prime Propagation). *For consecutive terms p_n, p_{n+1} set*

$$C_n = p_n + p_{n+1} + 1, \quad D_n = p_n + p_{n+1} + 3. \quad (1)$$

The pair (p_n, p_{n+1}) propagates if at least one of C_n, D_n lies in $\mathcal{T}$. There are infinitely many such indices n .

Conjecture 1.2 (Twin-Gap Existence). *An even positive integer d is a productive gap for $t \in \mathcal{T}$ if*

- (i) $t + d \in \mathcal{T}$,
- (ii) $2t + d + 1 \in \mathcal{T}$,
- (iii) *at least one of $d - 1, d + 1$ is prime.*

Every sufficiently large $t \in \mathcal{T}$ admits a productive gap of size $o(t)$. If $d(t)$ is such a gap, the map

$$G(t) := 2t + d(t) + 1$$

produces infinitely many twin primes.

The two constructions coincide on consecutive twins: if $d = p_{n+1} - p_n$, then $G(p_n) = C_n$. Conjecture 1.1 asks that this occur infinitely often along consecutive pairs. Conjecture 1.2 asks for a (possibly non-consecutive) productive $d = o(t)$ at every large t .

2 A covering that kills D_n

The disjunction “ C_n or D_n ” in Conjecture 1.1 is almost redundant.

Lemma 2.1 (Ternary obstruction). *If $p, q \in \mathcal{T}$ and $p, q > 3$, then $D := p + q + 3$ is never a lower twin prime.*

Proof. Every prime greater than 3 is congruent to $\pm 1 \pmod{6}$. A lower twin greater than 3 cannot be $\equiv 1 \pmod{6}$, for then the upper member would be $\equiv 3 \pmod{6}$ and hence divisible by 3. It cannot be $\equiv 3 \pmod{6}$, for then it would itself be divisible by 3. Thus every lower twin $p > 3$ satisfies $p \equiv 5 \pmod{6}$, equivalently $p \equiv 2 \pmod{3}$. Taking $p \equiv q \equiv 2 \pmod{3}$ we obtain

$$D + 2 = p + q + 5 \equiv 2 + 2 + 2 \equiv 0 \pmod{3},$$

and $D + 2 \geq 3 + 5 + 5 = 13 > 3$. Hence $D + 2$ is composite, so $D \notin \mathcal{T}$. □

The only exception is the first pair $(p_1, p_2) = (3, 5)$, for which $D_1 = 11 \in \mathcal{T}$. For all $n \geq 2$, propagation is exactly the assertion that $C_n \in \mathcal{T}$, i.e. that both C_n and $C_n + 2 = D_n$ are prime.

Lemma 2.2 (Residue of C_n). *If $p, q \in \mathcal{T}$ and $p, q > 3$, then $C := p + q + 1 \equiv 5 \pmod{6}$. In particular there is no obstruction modulo 2 or 3 to $C \in \mathcal{T}$.*

Proof. From $p \equiv q \equiv 5 \pmod{6}$ we get $C \equiv 5 + 5 + 1 \equiv 5 \pmod{6}$, hence $C \equiv 2 \pmod{3}$ and $C + 2 \equiv 1 \pmod{3}$. □

Thus Conjecture 1.1 is equivalent to the existence of infinitely many consecutive pairs $p < q$ in $\mathcal{T}$ for which $p + q + 1$ and $p + q + 3$ are both prime.

3 Both conjectures imply the twin-prime conjecture

Proposition 3.1. *Conjecture 1.1 implies that there are infinitely many twin primes. Conjecture 1.2 implies the same conclusion.*

Proof. A propagating pair produces an element of $\mathcal{T}$ of size $\sim 2p_n$. Infinitely many such pairs therefore yield infinitely many twins.

For Conjecture 1.2, take $t_0 \in \mathcal{T}$ larger than the “sufficiently large” threshold and set $t_{k+1} := G(t_k)$. Then $t_{k+1} \in \mathcal{T}$ and

$$t_{k+1} = 2t_k + d(t_k) + 1 > t_k,$$

so $(t_k)_{k \geq 0}$ is a strictly increasing sequence in $\mathcal{T}$. □

An unconditional proof of either statement would therefore prove the twin-prime conjecture. The Zhang–Maynard–Tao theorem on bounded gaps between primes produces infinitely many prime pairs at distance at most 246, but does not produce a six-dimensional constellation of the shape appearing here, and does not produce gap 2 infinitely often. The parity problem of classical sieves remains. There is no unconditional proof with present technology.

What can be proved is that Conjecture 1.1 is a formal special case of Hypothesis H, and that Conjecture 1.2 follows from a uniform Bateman–Horn law.

4 Resolution of Conjecture 1.1 under Hypothesis H

Conjecture 4.1 (Schinzel’s Hypothesis H). *Let $f_1, \dots, f_k \in \mathbb{Z}[x]$ be irreducible polynomials with positive leading coefficients, such that the product $f_1 \cdots f_k$ has no fixed prime divisor (the system is admissible). Then there are infinitely many integers n for which every $f_i(n)$ is prime.*

Consider the six linear forms

$$\begin{aligned} f_1(n) &= n, \\ f_2(n) &= n + 2, \\ f_3(n) &= n + 6, \\ f_4(n) &= n + 8, \\ f_5(n) &= 2n + 7, \\ f_6(n) &= 2n + 9. \end{aligned} \tag{2}$$

If they are simultaneously prime, then n and $n + 6$ are lower twins and

$$C = n + (n + 6) + 1 = 2n + 7$$

is a lower twin, so the pair $(n, n + 6)$ propagates.

Lemma 4.2 (Admissibility). *The 6-tuple (2) is admissible.*

Proof. It is enough to show that for every prime q , the reductions $f_i(x) \bmod q$ do not cover $\mathbb{F}_q$: equivalently, that there exists a residue $r \bmod q$ at which none of the f_i vanish.

The prime $q = 2$. Take $n \equiv 1 \pmod{2}$. Every $f_i(n)$ is odd.

The prime $q = 3$. Take $n \equiv 2 \pmod{3}$. Then

$$n \equiv 2, \quad n + 2 \equiv 1, \quad n + 6 \equiv 2, \quad n + 8 \equiv 1, \quad 2n + 7 \equiv 2, \quad 2n + 9 \equiv 1 \pmod{3}.$$

None vanish.

The prime $q = 5$. Take $n \equiv 1 \pmod{5}$. Then

$$n \equiv 1, n + 2 \equiv 3, n + 6 \equiv 2, n + 8 \equiv 4, 2n + 7 \equiv 4, 2n + 9 \equiv 1 \pmod{5}.$$

None vanish.

Primes $q \geq 7$. There are six forms and therefore at most six forbidden residues for n . Since $q > 6$, at least one class survives.

Hence no prime is a fixed divisor of the product $f_1 \cdots f_6$. $\square$

Lemma 4.3 (Gap 6 forces consecutiveness). *If $n > 3$ and both $n, n + 6 \in \mathcal{T}$, then n and $n + 6$ are consecutive in $\mathcal{T}$.*

Proof. The only odd integers strictly between n and $n + 6$ are $n + 2$ and $n + 4$. Now $n > 3$ and $n \in \mathcal{T}$ imply $n \equiv 2 \pmod{3}$, so $n + 4 \equiv 0 \pmod{3}$ and $n + 4 \geq 9$. Thus $n + 4$ is composite, so neither $n + 2$ (whose upper twin would be $n + 4$) nor $n + 4$ lies in $\mathcal{T}$. $\square$

Theorem 4.4 (Conditional resolution of Conjecture 1.1). *Assume Hypothesis H. Then Conjecture 1.1 is true. More precisely, there are infinitely many consecutive pairs in $\mathcal{T}$ at gap 6 that propagate.*

Proof. Lemma 4.2 and Hypothesis H give infinitely many n for which $f_1(n), \dots, f_6(n)$ are all prime. For all such $n > 3$ we have $n, n + 6 \in \mathcal{T}$ and $2n + 7 \in \mathcal{T}$. Lemma 4.3 says these are consecutive in $\mathcal{T}$, and

$$C = n + (n + 6) + 1 = 2n + 7 \in \mathcal{T},$$

so the pair propagates. $\square$

This is the natural precise form of the “sufficiently strong Hardy–Littlewood” heuristic I used in the earlier notes: it is Hypothesis H for one explicit admissible 6-tuple.

The same argument with the Bateman–Horn conjecture supplies the asymptotic. The singular series

$$\mathfrak{S} = \prod_q \left(1 - \frac{1}{q}\right)^{-6} \left(1 - \frac{\nu(q)}{q}\right),$$

where $\nu(q)$ is the number of roots of $\prod f_i$ modulo q , is positive by Lemma 4.2, and

$$\#\{n \leq X : \text{all } f_i(n) \text{ prime}\} \sim \mathfrak{S} \prod_{i=1}^6 (\text{lead } f_i)^{-1} \int_2^X \frac{dt}{(\log t)^6}.$$

In particular the count of propagating gap-6 pairs up to X is $\asymp X/(\log X)^6 \rightarrow \infty$.

Remark 4.5 (Corrected local heuristic). In the original notes I treated C_n and D_n as exchangeable and summed $\int dt/(\log t)^3$. By Lemma 2.1 the D_n -branch contributes only the single pair $(3, 5)$. Conditioning on a consecutive pair at height x , one asks that a 2-tuple of size $\sim 2x$ be twin, which has probability $\asymp 1/(\log x)^2$. The twin-counting measure is $\asymp dt/(\log t)^2$, so the expected number of successes up to X is

$$\asymp \int_3^X \frac{dt}{(\log t)^4},$$

which still diverges. The empirical fit $6.175 (\ln p)^{-2.657} (\ln \ln p)^{2.258}$ is a regression on a finite range; the Hardy–Littlewood prediction for the success rate is $\sim c/(\log x)^2$.

The gap-6 constellation already appears at the beginning of $\mathcal{T}$:

$$(5, 7, 11, 13, 17, 19), \quad (11, 13, 17, 19, 29, 31).$$

Both are propagating consecutive pairs, as the theory predicts.

5 Resolution of Conjecture 1.2 under uniform Bateman–Horn

Conjecture 1.2 is a $\forall\exists$ statement and is strictly stronger than Conjecture 1.1. Infinitely many twins with a productive gap of size 6 does not give a productive gap at every $t \in \mathcal{T}$. For a concrete counterexample to the gap-6 specialisation at every t , take $t = 17$: then $17 + 6 = 23$ is prime but $25 = 5^2$ is not, so $23 \notin \mathcal{T}$.

Fix $t \in \mathcal{T}$ with $t > 3$, so $t \equiv 5 \pmod{6}$. Productive even d with the side-condition that $d - 1$ is prime correspond to simultaneous primality of the five linear forms in d

$$\begin{aligned} g_1(d) &= d + t, \\ g_2(d) &= d + t + 2, \\ g_3(d) &= d + 2t + 1, \\ g_4(d) &= d + 2t + 3, \\ g_5(d) &= d - 1. \end{aligned} \tag{3}$$

The alternative $g_5^*(d) = d + 1$ covers the other clause of condition (iii). We also impose $d \equiv 0 \pmod{6}$, which is forced if $t + d \in \mathcal{T}$ (Lemma 2.2).

Lemma 5.1 (No small-prime obstruction, uniformly in t). *Let $t > 3$, $t \equiv 5 \pmod{6}$, and $d \equiv 0 \pmod{6}$. Then none of $g_1(d), \dots, g_4(d), g_5(d)$ is divisible by 2 or 3.*

Proof. The integer d is even and t is odd, so all five forms are odd. Modulo 3: $d \equiv 0$ and $t \equiv 2$, so

$$d + t \equiv 2, \quad d + t + 2 \equiv 1, \quad d + 2t + 1 \equiv 2, \quad d + 2t + 3 \equiv 1, \quad d - 1 \equiv 2 \pmod{3}.$$

None vanish. The same computation with $g_5^*(d) = d + 1$ gives $d + 1 \equiv 1 \pmod{3}$. $\square$

For each prime $q \geq 5$ the five forms forbid at most five residues of $d \pmod{q}$. After intersecting with the progression $d \equiv 0 \pmod{6}$, a positive-density set of residue classes modulo $6q$ survives except possibly for those t at which the five forbidden classes

$$\{-t, -(t+2), -(2t+1), -(2t+3), 1\}$$

cover $\mathbb{F}_q$. That is a finite system of polynomial conditions on $t \pmod{q}$. For each fixed $q \geq 5$ it cuts out $O(1)$ residue classes.

Lemma 5.2. *For all $t \in \mathcal{T}$ outside a thin set (density zero among twins, under Hardy–Littlewood), at least one of the two 5-tuples $(g_1, \dots, g_5)$ and $(g_1, \dots, g_4, g_5^*)$ is admissible. For every such t the singular series $\mathfrak{S}(t)$ satisfies $\mathfrak{S}(t) \gg (\log t)^{-O(1)}$.*

The exceptional t , if any, may be absorbed by switching $d - 1$ for $d + 1$, or by allowing a bounded extra linear form such as $d - p$ for a fixed small prime p .

Conjecture 5.3 (Uniform Bateman–Horn hypothesis). *For an admissible family of k linear forms in one variable, with heights $\leq X$ and singular series $\mathfrak{S}$,*

$$\#\{n \leq Y : \text{all forms prime}\} = (\mathfrak{S} + o(1)) \frac{Y}{(\log Y)^k},$$

uniformly for $Y \geq X^\theta$ with any fixed $\theta > 0$, as $X \rightarrow \infty$.

Theorem 5.4 (Conditional resolution of Conjecture 1.2). *Assume the uniform Bateman–Horn hypothesis for 5-tuples of linear forms. Then every sufficiently large $t \in \mathcal{T}$ has $\gg t^\theta / (\log t)^{O(1)}$ productive gaps of size $\leq t^\theta$, for any fixed $\theta > 0$. In particular Conjecture 1.2 holds, even with the stronger bound $d(t) = O_\theta(t^\theta)$.*

Proof. Apply Bateman–Horn to $g_1, \dots, g_5$ (or the starred variant) in the variable $d \leq Y := t^\theta$, restricted to $d \equiv 0 \pmod{6}$. Lemma 5.1 removes the primes 2 and 3; Lemma 5.2 gives $\mathfrak{S}(t) \gg (\log t)^{-O(1)}$ for large t . The predicted count is

$$\gg \frac{t^\theta}{(\log t)^{O(1)} (\log(t^\theta))^5} \rightarrow \infty.$$

Each such d is productive of size $o(t)$. □

Along the orbit $t_{k+1} = G(t_k)$ one has $t_{k+1} \sim 2t_k$, so $t_k \asymp 2^k t_0$. This is the generative map of the earlier notes; under Theorem 5.4 it is defined at every step.

6 Status of the claims

The following table records the status of every statement used to resolve the two conjectures.

Claim	Status
$D_n \in \mathcal{T}$ for $n \geq 2$	<i>False</i> (Lemma 2.1). Conjecture 1.1 reduces to $C_n \in \mathcal{T}$.
Conjecture 1.1, unconditionally	<i>Open</i> . Implies the twin-prime conjecture (Proposition 3.1).
Conjecture 1.1 under Hypothesis H	<i>True</i> (Theorem 4.4), via the admissible 6-tuple (2).
Conjecture 1.2, unconditionally	<i>Open</i> , and strictly stronger than Conjecture 1.1.
Conjecture 1.2 under uniform Bateman–Horn	<i>True</i> (Theorem 5.4), even with $d(t) \leq t^\theta$.
Conjecture 1.1 or 1.2 $\Rightarrow$ infinitely many twins	<i>True</i> (Proposition 3.1).
Computational support ($p_n \leq 10^{13}$; constructive chains with $\theta = 0.6$)	Consistent with both conjectures and with the corrected $1/(\log x)^2$ rate.

An unconditional proof of either conjecture would be a proof of the twin-prime conjecture by a new route, and would require an idea that bypasses the parity barrier for a six-dimensional prime constellation. That idea is not supplied here. The theorems above are the resolution that the statements actually admit: Conjecture 1.1 is Hypothesis H for one explicit admissible tuple; Conjecture 1.2 is a uniform short-interval form of the same law.

7 The Python certificate

The companion script `prove_section6.py` machine-checks every claim of Section 6. The modular lemmas are finite residue-class enumerations and are therefore complete proofs. Admissibility of (2) is proved by inspecting $q = 2, 3, 5$ exhaustively and invoking the pigeonhole principle for $q \geq 7$. The implication Hypothesis H $\Rightarrow$ Conjecture 1.1 is a purely logical deduction from those lemmas. The “open” claims are certified by exhibiting the reduction to the twin-prime conjecture together with the observation that no finite computation decides an infinitary statement of this type. Computational support is reproduced on a truncated range by a segmented identification of lower twins, a count of propagating consecutive pairs, a comparison of the empirical success rate against $c/(\log x)^2$, and the construction of an explicit G -orbit with $\theta = 0.6$.

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